HDOC- 11
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Program to Display the Month Name and Number of Days Using switch-case

This document provides a clean, well-commented implementation for displaying the name of a month and its number of days from a month number in C. The program takes a number between 1 and 12 as input and uses a switch-case statement to find the matching month name and number of days.

1. C Implementation

Step 1: Understand the Logic

- The numbers 1 to 12 are mapped to the months January to December.
- Months with 31 days: January, March, May, July, August, October, December.
- Months with 30 days: April, June, September, November.
- February has 28 days (29 days in a leap year). This program uses 28.
- Each case of the switch sets the month name and the number of days, and ends with break so the next case does not run.
- The default case handles any number outside 1 to 12 and ends the program with an error message.

Step 2: Define Variables and Data Types

- Use int variables for the month number (month) and the number of days (days).
- Use a character pointer (char *name) to hold the month name.

Step 3: Take User Input

- Display a prompt using printf("Enter a month number (1-12): "), then read the value using scanf("%d", &month).

Step 4: Apply the switch-case Statement

1. Write switch (month) so that the value of month is compared with each case.
2. For case 1 to case 12, store the month name and the number of days, then break.
3. In the default case, print an error message and stop the program.

Step 5: Format the Output

- Print the month name and the number of days using printf().

Complete C Code

```c
/*
 * Q16 (User Inputs, Operations & Output)
 * Write a program to display the month name and number of days
 * using switch-case for a given month number.
 */
#include <stdio.h>

int main() {
    int month, days;
    char *name;

    // Step 1: Read the month number
    printf("Enter a month number (1-12): ");
    scanf("%d", &month);

    // Step 2: Find the month name and number of days using switch-case
    switch (month) {
        case 1:  name = "January";      days = 31; break;
        case 2:  name = "February";     days = 28; break;   // 29 if leap
        case 3:  name = "March";        days = 31; break;
        case 4:  name = "April";        days = 30; break;
        case 5:  name = "May";          days = 31; break;
        case 6:  name = "June";         days = 30; break;
        case 7:  name = "July";         days = 31; break;
        case 8:  name = "August";       days = 31; break;
        case 9:  name = "September";    days = 30; break;
        case 10: name = "October";      days = 31; break;
        case 11: name = "November";     days = 30; break;
        case 12: name = "December";     days = 31; break;
        default:
            printf("Invalid input! Enter a number between 1 and 12.\n");
            return 0;
    }

    // Step 3: Display the result
    printf("Month: %s\n", name);
    printf("Number of days: %d\n", days);

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc11.c -o hdoc11
$ ./hdoc11
Enter a month number (1-12): 1
Month: January
Number of days: 31
```

Run 2:

```
$ ./hdoc11
Enter a month number (1-12): 2
Month: February
Number of days: 28
```

Run 3:

```
$ ./hdoc11
Enter a month number (1-12): 4
Month: April
Number of days: 30
```

Run 4:

```
$ ./hdoc11
Enter a month number (1-12): 13
Invalid input! Enter a number between 1 and 12.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 1
- Calculation:
  - o switch (month) compares month = 1 with the case labels
  - o It matches case 1, so name = "January" and days = 31
  - o break exits the switch, then the name and days are printed
- Output: Month: January, Number of days: 31

Sample Test Case 2:

- Input: 2
- Calculation:
  - o switch (month) compares month = 2 with the case labels
  - o Case 1 does not match; it matches case 2, so name = "February" and days = 28
  - o break exits the switch, then the name and days are printed
- Output: Month: February, Number of days: 28

Sample Test Case 3:

- Input: 4
- Calculation:
  - o switch (month) compares month = 4 with the case labels
  - o Cases 1, 2 and 3 do not match; it matches case 4, so name = "April" and days = 30
  - o break exits the switch, then the name and days are printed
- Output: Month: April, Number of days: 30

Sample Test Case 4:

- Input: 13
- Calculation:
  - o switch (month) compares month = 13 with the case labels
  - o None of the cases 1 to 12 match 13
  - o The default case prints the error message and return 0 ends the program, so no month is printed
- Output: Invalid input! Enter a number between 1 and 12.